

Is My Division Answer Reasonable?

A quick-check skill for every division problem — Chusaki Math Coach

Why this matters: Every time you divide, a small slip — a dropped zero, a decimal in the wrong spot, a remainder that's too big — can turn a correct method into a wrong answer. Before you write your final answer, a 5-second reasonableness check catches these slips WITHOUT redoing the whole problem. This is the same habit behind the "Finish-Line Check" you already use on every test.

The Big Idea: Two Quick Checks

- 1. Estimate first.** Round the number you're dividing (the dividend) to a friendly number, then divide that in your head. Your real answer should land close to this estimate.
- 2. Multiply back.** divisor $\times$ quotient (+ remainder, if there is one) should give you back the number you started with. This is the fastest way to catch a wrong answer.

When an Answer IS Reasonable

- The method follows the normal division steps — nothing was skipped.
- The size of the answer makes sense: dividing by a number bigger than 1 should give a SMALLER result than you started with.
- Multiplying back (divisor $\times$ quotient + remainder) gets you back to the start number.

When an Answer is NOT Reasonable

- The answer is BIGGER than the number you started with (when dividing by a normal whole number greater than 1).
- A small mistake (like a dropped zero or a shifted decimal point) made the number way too large or way too small.
- The remainder is BIGGER than (or equal to) the divisor — this is always a sign something went wrong.

The Remainder Rule (the #1 exam trap): the remainder must ALWAYS be smaller than the divisor. If you divide by 6, your remainder can only be 0, 1, 2, 3, 4, or 5 — never 6 or more. A remainder of 6+ when dividing by 6 means you can still divide at least once more.

Worked Examples

Problem & Answer Given	Reasonable?	Why
$438 \div 6 = 73$	YES	Estimate: $420 \div 6 = 70$, close to 73. Check: $6 \times 73 = 438$. Matches exactly.
$47 \div 5 = 8 \text{ remainder } 9$	NO	Remainder (9) is BIGGER than the divisor (5) — broke the remainder rule. Correct: 9 remainder 2.

$84 \div 4 = 336$	NO	The answer (336) is bigger than the start number (84). Dividing by 4 should make it smaller. Correct: 21.
$96 \div 3 = 3.2$	NO	Estimate: $90 \div 3 = 30$, but the answer is only 3.2 — way too small (a dropped zero). Correct: 32.

Your Turn — 4 Questions

For each one: is the answer reasonable? If not, say why (too big, too small, or remainder rule) and find the correct answer.

P1. $156 \div 3 = 520$ — reasonable?

P2. $74 \div 8 = 9$ remainder 2 — reasonable?

P3. $53 \div 6 = 7$ remainder 11 — reasonable?

P4. Estimate first (round 364 to 350): $364 \div 7 = 52$ — reasonable?

Answer Key

P1: NOT reasonable — 520 is bigger than the start number (156). Correct: 52.

P2: Reasonable — $8 \times 9 + 2 = 74$. ✓ Matches, and remainder 2 < divisor 8.

P3: NOT reasonable — remainder (11) is bigger than the divisor (6), breaks the remainder rule.
Correct: 8 remainder 5.

P4: Reasonable — estimate ($350 \div 7 = 50$) is close to the real answer (52).

Remember this every time you divide: estimate first so you know roughly what to expect, multiply back to check your exact answer, and always check the remainder is smaller than the divisor. Three checks, five seconds, one less careless mistake on test day.